

Dataset Demographic and Characteristics

TABLE S1. DEMOGRAPHIC, AND LABORATORY CHARACTERISTICS OF THE STUDY COHORT. DATA ARE PRESENTED AS MEDIAN (IQR) FOR CONTINUOUS VARIABLES AND N (%) FOR CATEGORICAL VARIABLES.

Characteristic	1D displacement-encoded Overall cohort (N=159)	3D displacement-encoded Overall cohort (N=50)
Demographic data		
Age, years	43.0 (31.5–52.0)	45.0 (33.0–56.8)
Male sex, n (%)	64 (40.3%)	20 (40.0%)
Female sex, n (%)	95 (59.7%)	30 (60.0%)
Weight, kg	78.2 (67.1–92.8)	79.0 (68.4–85.3)
Height, cm	166.2 (160.9–175.0)	167.5 (161.0–179.0)
BMI, kg/m ²	27.1 (24.2–32.5)	26.5 (23.3–31.5)
Race, n (%)		
White	104 (65.4%)	34 (68.0%)
Black/African American	28 (17.6%)	6 (12.0%)
Asian	13 (8.2%)	5 (10.0%)
American Indian/Alaska Native	1 (0.6%)	2 (4.0%)
Multiple race	5 (3.1%)	---
Other	1 (0.6%)	2 (4.0%)
Unknown	7 (4.4%)	1 (2.0%)
Diagnosis, n (%)		
STAT3 hyper-IgE syndrome	45 (28.3%)	---
NASH	26 (16.4%)	11 (22.0%)
Lipodystrophy	14 (8.8%)	8 (16.0%)
Noncirrhotic portal hypertension	10 (6.3%)	8 (16.0%)
SOCS1	10 (6.3%)	3 (6.0%)
Keto healthy volunteer	9 (5.7%)	4 (8.0%)
Obesity phenotyping	10 (6.3%)	---
Alcohol use disorder	9 (5.7%)	1 (2.0%)
Hep B	4 (2.5%)	3 (6.0%)
CTLA4	4 (2.5%)	3 (6.0%)
Androgen insensitivity syndrome	4 (2.5%)	3 (6.0%)
Cushings	6 (3.8%)	---
Healthy volunteer	3 (1.9%)	---
Nodular regenerative hyperplasia	2 (1.3%)	---

CVID	---	2 (4.0%)
CGD	---	1 (2.0%)
Other	3 (1.9%)	3 (6.0%)
Laboratory data		
Alkaline phosphatase, U/L	75.0 (56.0–86.2)	68.5 (56.0–89.8)
ALT, U/L	25.0 (18.0–45.5)	28.5 (17.8–43.8)
AST, U/L	24.0 (20.0–36.8)	25.0 (21.0–36.0)
Total bilirubin, mg/dL	0.50 (0.30–0.70)	0.50 (0.40–0.72)
Direct bilirubin, mg/dL	0.20 (0.10–0.20)	0.20 (0.10–0.30)

Abbreviations: BMI, body mass index; ALT, alanine aminotransferase; AST, aspartate aminotransferase; IQR, interquartile range. Percentages may not sum to 100 due to rounding. Categorical proportions are computed over the column N. Lab-value missingness: non-numeric entries (e.g., "HEMOLYZED") were treated as missing for summary statistics.